

What the sorter actually does, and what the neurons actually do

A ground-truth audit of Kilosort4, seven modifications to it, and two post-hoc correction tools — followed by the first behavioural physiology from session 20260916_110311.

2 OCT 2026 • 52 PLACEMENTS × 4 REPLICATES × 9 CONFIGURATIONS • 478 COMPLETED TRIALS • 111 HAND-CURATED UNITS

95% **Detection is near-perfect.** Of every injected spike, 94–96% lands in some cluster. Kilosort almost never misses a spike outright.

40% **Assignment is where it fails.** On an average cluster, 40% of the spikes belong to a different neuron.

0 of 16 **Merging doesn't fix it.** Combining a unit's split pieces trades down in all 16 algorithm × tier cells tested.

2.3% **Post-hoc correction doesn't fix it either.** Of 11,027 spikes it recovered, 259 genuinely belonged to the unit.

93.5% **One cause behind all of it.** What a unit's template matches on its own channel is overwhelmingly a *neighbour's* real spike.

How the test data was made

None of this could be measured on real recordings, because on real data nobody knows which neuron a spike came from. So we built recordings where we *do* know — real voltage with real neurons in it, plus a handful of extra neurons whose every spike time we chose ourselves.

The donor neurons

You hand-reviewed every candidate in the review tool. **33 units labelled good** survived as donors, after also requiring low contamination, a compact spatial footprint, and — for the pair tier — a cross-correlogram refractory test confirming two donors are genuinely different cells rather than one cell Kilosort had split.

Crucially we inject **real recorded spike snippets**, not copies of a template. A template copy would be identical every time, which flatters any algorithm: real spikes vary in amplitude and shape from firing to firing, and that variability is exactly what makes sorting hard.

The four difficulty tiers

Every site on the probe was ranked by **the amplitude of the largest neuron already living there** — your correction to the original design, which had ranked sites by counting events instead. That measure spans 23–467 μV across this probe, a 20 $\times$ contrast to build tiers from.

TIER	PER REC.	WHERE IT GOES	WHAT MAKES IT HARD
easy	3	quiet site, 23–70 μV neighbours	nothing — the control
collision	3	quiet site	each spike placed 4–25 samples (0.13–0.83 ms) from a real resident neuron's spike, so the two waveforms physically overlap
pair	2 pairs	quiet sites, 3 channels apart	two donors placed as near-neighbours — must be told apart from each other
hard	3	loud site, 223–466 μV neighbours	a large neighbour dominates the channel

13 placements per recording, 4 recordings from different windows of the session, ~290 spikes each — 52 placements and 15,086 ground-truth spikes in total.

Two design points worth stating, both of which came out of your review:

- **Collision and pair sit on quiet sites deliberately.** Their challenge is separating units, so channel noise must not be confounded with it. Only the hard tier gets loud sites.

- **The pair donors are remote in the real recording** — 181–242 μm apart with template similarity 0.000. If they had been near-neighbours they might have been one neuron Kilosort split, which would invert what the tier measures.

Injection times avoid the resident neurons' own spikes by a 75-sample guard (except in the collision tier, where the overlap is the point), each unit's tier **rotates across the four recordings** so tier differences are measured within unit rather than across units, and one configuration is simply **stock Kilosort run twice** — the noise floor, which is how we know the hard tier is unreliable.

What each algorithm changed

Seven modifications, each a single flag-gated patch swapped in at runtime. The installed Kilosort package is never modified, and turning a flag off reproduces stock Kilosort exactly.

NAME	WHAT IT DOES	THE REASONING
vanilla	nothing — stock Kilosort4	the baseline
vanilla repeat	nothing — stock, run a second time	measures the noise floor, since Kilosort is not fully deterministic
subsample align	Corrects each spike's timing to a fraction of a sample before its features are computed, using the wavelet phase estimator.	Kilosort snips every spike at an integer sample, so up to half a sample of timing error leaks straight into the features the clustering runs on.
coarse + fine timing	Slides the snippet ± 6 whole samples against the average spike shape <i>first</i> , then applies the sub-sample correction.	A fix for a measured bug in the above: 55% of spikes are more than half a sample off, so the phase stage was being handed offsets outside the range where it is even meaningful.
amplitude normalisation	Scales every spike's feature vector to unit length, removing amplitude as a dimension.	Amplitude is the largest-variance direction within one neuron's spikes, and Kilosort seeds 200 clusters per region with k-means — which will cut along exactly that direction, slicing one neuron into several.

NAME	WHAT IT DOES	THE REASONING
timing + amplitude	Both of the above together.	tests whether the two combine
coarse + timing + amplitude	All three together.	the full stack
footprint ×1	Appends each spike's spatial footprint — the normalised pattern of amplitude across nearby channels — to the features the clustering sees, weighted equal to the waveform features.	Two spikes from one neuron must share a footprint; two neurons sitting close together need not. Footprint separates a hard pair at AUC 0.999 where a single channel manages 0.749.
footprint ×3	The same, weighted three times as strongly so footprint dominates.	The weight <i>is</i> the question — if the idea works at all, the two weights should differ systematically rather than randomly.

All seven target one of two stages: where spike snippets are measured (detection) or what the clustering is allowed to see.

Note what is and isn't being changed. The footprint patches alter only *what the clustering sees*, never the clustering algorithm itself. The timing patches alter only *where snippets are measured*. Every one is reversible, which is what makes the comparison against stock trustworthy.

One correction to the working assumption

“About 50 percent of the detected events in easy channels do not belong to the real unit, even though most spikes are detected.”

The second half is right. The first half was true of the **first** benchmark and is no longer. That ~50% came from round one, where “quiet” channels were picked by counting events rather than by amplitude — and those channels still hosted 5–11 neurons. After you corrected the criterion to *neighbour amplitude*, the v2 easy tier improved substantially:

BENCHMARK	RECALL	PRECISION	FOREIGN
Round 1 (event-count “quiet” sites)	91.3%	53.8%	46%

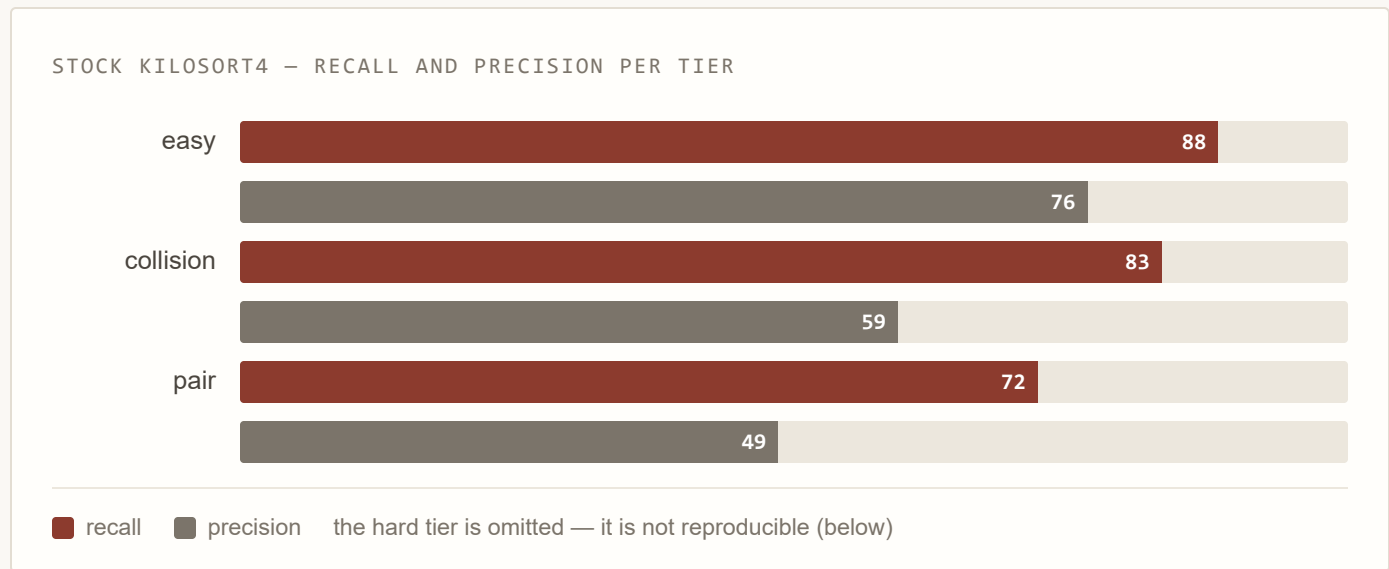
BENCHMARK	RECALL	PRECISION	FOREIGN
Round 2 (amplitude-chosen quiet sites)	88.3%	76.5%	24%

Stock Kilosort, easy tier. “Foreign” = spikes in the matched cluster that are not the injected unit’s.

So on genuinely quiet channels about **a quarter** of a cluster is foreign, not half. The ~50% figure still describes two real things: the **pair tier** (49% foreign) and roughly the **session-wide average** (40% foreign). And the foreign spikes are not noise — **80.6% are other real neurons** firing on the same channel, against a 3.9% chance rate.

How Kilosort performs, by difficulty

Recall = of the unit’s spikes, what share landed in its best-matching cluster. **Precision** = of that cluster’s spikes, what share are really the unit’s. Both always reported together.



Recall holds up reasonably everywhere. Precision is the problem, and it collapses exactly where neurons sit close together: **on the pair tier, half of what Kilosort calls one neuron is another neuron.**

The hard tier is not measurable. Two bit-identical runs of the same code disagree on 2 of its 12 placements by up to 0.444 recall. Kilosort is non-deterministic there (and only there), so no hard-tier number in this project should be quoted. Easy, collision and pair reproduce exactly.

Seven modifications, one real winner

MODIFICATION	Δ RECALL	Δ PRECISION	VERDICT
Footprint-aware clustering ×3	+1.8	+2.2	pair recall 72→80, p=0.038
Footprint-aware clustering ×1	-2.3	+5.4	precision 60→65, p=0.010
Coarse + fine timing	-2.3	-1.6	no effect
Sub-sample alignment	-2.8	-2.8	no effect
Timing + amplitude norm.	-2.3	+0.6	no effect
Amplitude normalisation	-3.4	+2.6	worse (p=0.012)
Coarse + timing + amplitude	-3.6	+2.3	worse (p=0.023)

Paired Wilcoxon on the same 52 placements against stock. Only effects clearing the measured noise floor are shown as wins.

Five of seven do nothing or hurt. The two that work are the two that use **spatial footprint** — how a spike's voltage spreads across neighbouring channels. These are the only statistically supported improvements this project has produced, and they help precisely where the failure is: crowded neighbourhoods.

Why merging the pieces cannot fix it

The natural hypothesis: Kilosort splits a neuron into several clean clusters, so pool them. Tested on all eight configurations. **The premise doesn't hold.**

- Units average **1.27–1.38** substantial clusters, never more than 3, and **67–77% have exactly one**. There is no three-way split to merge.
- Where a second piece exists, pooling trades down in **all 16** algorithm × tier cells: **+3 to +7 points of recall for -4 to -22 of precision**.
- A perfect merge tool converges every algorithm to roughly the same 77% recall / 46% precision — merging cannot even separate them.

The reason is in the shape of the loss. The neuron is not broken into clean thirds. It is **one dominant cluster plus a long thin smear** — 14 to 22 spikes per 100 scattered a handful at a time across many clusters, each too small to count as a piece, and all of them full of other neurons. Pooling imports that contamination.

Why post-hoc correction cannot fix it either

The pipeline built earlier in this project — converge the template by wavelet realignment, whiten with the noise precision matrix, then matched-filter the whole session for missed spikes — had been run twice on real data, and both times the headline had to be withdrawn, because on real data you cannot check whether a recovered spike belongs to the unit. The synthetic benchmark removes that limitation.

TIER	ACCEPTED	REALLY THE UNIT'S	RECALL	PRECISION
easy	144	4	.883 → .897	.765 → .739
collision	208	4	.832 → .845	.594 → .539
pair	334	5	.720 → .739	.486 → .440
pooled	11,027	259 (2.3%)	+1.7	-4.3

Stock Kilosort, 52 placements, truth-blind pipeline. Chance control: same accepted set against random times.

It is not finding noise — chance gives 0.1%, so 2.3% is 23× above chance and the matched filter genuinely works. The problem is *what* it finds:

Of the 10,768 wrongly accepted spikes, **10,070 (93.5%) coincide with a spike Kilosort had already filed under a different cluster.** The tool recovers real spikes belonging to other neurons.

What kind of spike was recovered

ALGORITHM	MISFILED	UNDETECTED	RECOVERED MISFILED	RECOVERED UNDETECTED
Original Kilosort	2,480 (79%)	642 (21%)	212 (8.5%)	47 (7.3%)

ALGORITHM	MISFILED	UNDETECTED	RECOVERED MISFILED	RECOVERED UNDETECTED
Footprint ×3	2,085 (73%)	767 (27%)	262 (12.6%)	25 (3.3%)
Coarse + fine timing	2,625 (76%)	848 (24%)	208 (7.9%)	34 (4.0%)

Every spike absent from a unit's best cluster is either misfiled (detected, wrong cluster) or undetected (nobody has it).

So it does both: of its 259 correct recoveries, **82% were spikes taken back from another unit** and 18% were genuinely new detections. But it reaches only ~8% of either pool. Two things here matter more than the headline:

- The **79/21 split independently reproduces** the earlier finding that 84% of failures are a clustering failure, not a detection failure — different data, different method, same answer.
- **Footprint ×3 recovers more misfiled spikes (12.6% vs 8.5%) and fewer undetected ones (3.3% vs 7.3%)** — exactly the signature of a method better at *attribution* and no better at *detection*.

The merge detector was also tested against a known answer for the first time, and scored **0.00–0.11 sensitivity**. The reason is diagnosable: the refractory violation rate of a truly-same-neuron merge (0.066–0.076) is barely below that of genuinely different neurons (0.076–0.123), because the input clusters already violate at 0.051–0.060 on their own. The test needs clean inputs and does not get them.

Where this leaves the sorting problem

Not detection — 95% of spikes are found. Not splitting — most units are not split. Not merging, and not post-hoc recovery; both buy a couple of points of recall and cost four to ten of precision. Every route fails at the same wall:

Anything that matches a unit's template on its own channel is overwhelmingly a neighbour's real spike. This is a per-spike *attribution* problem — deciding which of several nearby neurons each spike came from.

Three independent measurements point at the same instrument. Footprint similarity separates two same-channel neurons at AUC **0.999** where the single-channel score manages 0.749. The only two statistically supported wins are both footprint configurations. And the footprint gate

roughly doubles recovery precision at every threshold. In all three it was used as a clustering *weight* or an after-the-fact *filter* — never as the thing that decides ownership. **A footprint-first search, scoring each event against every nearby unit and assigning it to the best, has not been tried.** That is the open lead.

The physiology: session 20260916_110311

Sync pipeline run on the full session. Quality is not in question: **21,422 / 21,422** Bpod events matched, local-interpolation held-out error **median 0.000 ms, max 43 ms**, no gaps, all 953 trials valid. The pipeline's ephys-offset fix engaged and recovered **87.51 s** — without it every spike would have been assigned to the wrong trial.

Trials and units. 478 of 953 trials are completed and sync-valid (50%); the rest are genuine incompletes — never poked centre, broke fixation, withdrew early, or never chose. Units are the **111 you labelled good in the review tool**. Note `units.parquet`'s own `quality_label` column is Kilosort's KSLabel and would have given 189 units — not the curation you asked for.

EVENT	SIGNIFICANT	SUBSTANTIAL	UP / DOWN	SIDE-SELECTIVE
End of cue stimulus	54 / 111	44 (40%)	27 / 27	45 (41%)
Reward delivery	30 / 111	21 (19%)	21 / 9	15 (14%)
Give-up, no water (control)	13 / 111	10 (9%)	6 / 7	18 (16%)

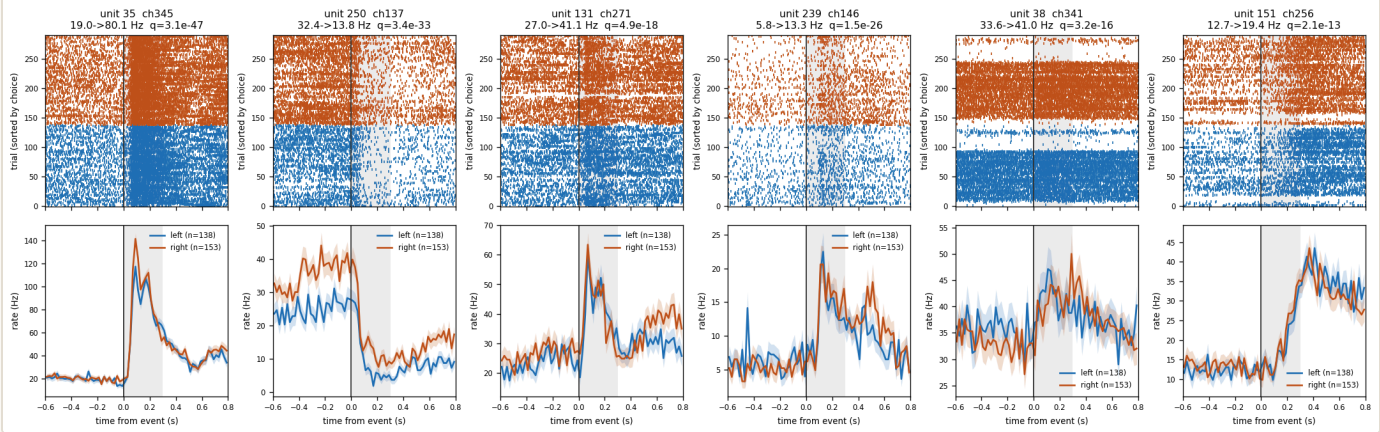
“Substantial” = $FDR < 0.05$ and $|modulation| \geq 0.10$. With 478 paired trials significance alone detects very small shifts, so the substantial column is the one to quote.

Reward

Aligned to the valve opening (`water_L` / `water_R`), 291 trials. Responses are large and tightly time-locked — unit 35 goes **19 → 80 Hz**, unit 250 drops **32 → 14 Hz**, unit 131 rises 27 → 41 Hz. Onsets sit on the valve instant, not on the choice poke a median 1.61 s earlier.

They also look **reward-specific**: all 21 substantially responsive units were re-tested at the give-up moment with identical windows. **18 show nothing there, and the other 3 respond in the opposite direction.** None responds the same way.

Session 20260916_110311 — strongest responses at reward delivery (valve open)
completed trials only, review-tool 'good' units, split by choice side

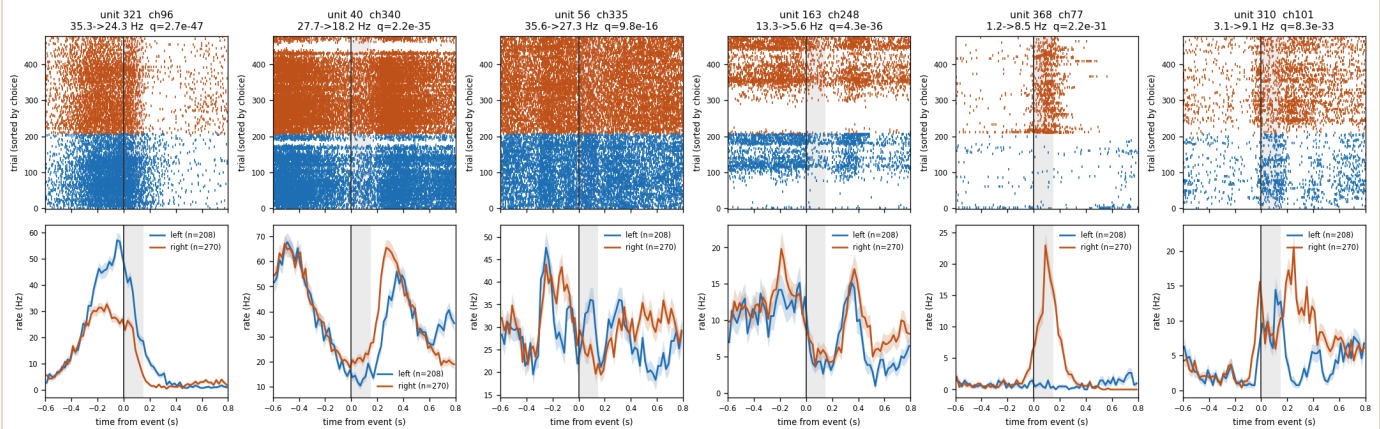


Six strongest reward responses. Rasters sorted by choice side (blue = left, orange = right); shaded band is the 0–300 ms test window. Note how sharply unit 35's onset locks to valve open.

End of cue

Aligned to the end of `stimulus_delivery` — verified identical to the start of `wait_sin` on all 479 trials, since this session's stimulus time and minimum sampling time are both 0.35 s. More units respond here than at reward, and far more carry **side information**: 45 of 111 differ between left and right choices, against 15 at reward. Unit 368 is the clearest case — **0.6 Hz on left-choice trials, 14.5 Hz on right**, from a 1.2 Hz baseline.

Session 20260916_110311 — strongest responses at end of cue stimulus
completed trials only, review-tool 'good' units, split by choice side



Six strongest cue-end responses. Unit 368 (fifth panel) fires almost exclusively on right-choice trials. Unit 321 shows a left-prefering ramp that peaks just before cue offset.

Overall: **32 units respond at cue end only, 8 at reward only, 22 at both, 49 at neither.** Side is carried at the choice point, not at the outcome.

Two limits worth stating plainly

- **Cue offset and movement onset are the same instant in this task.** Movement time is measured from `wait_sin`'s start and its median is 0.253 s. The test window was cut to 150 ms to close before the 10th-percentile movement time, but *no* window can separate “responds to the stimulus ending” from “responds to initiating a choice” given this design.
- **The give-up moment is not a passive omission marker** — it is the rat's own behavioural report (see below), since no *delivered* negative outcome exists at a matched time. So the specificity result above is suggestive, not decisive.

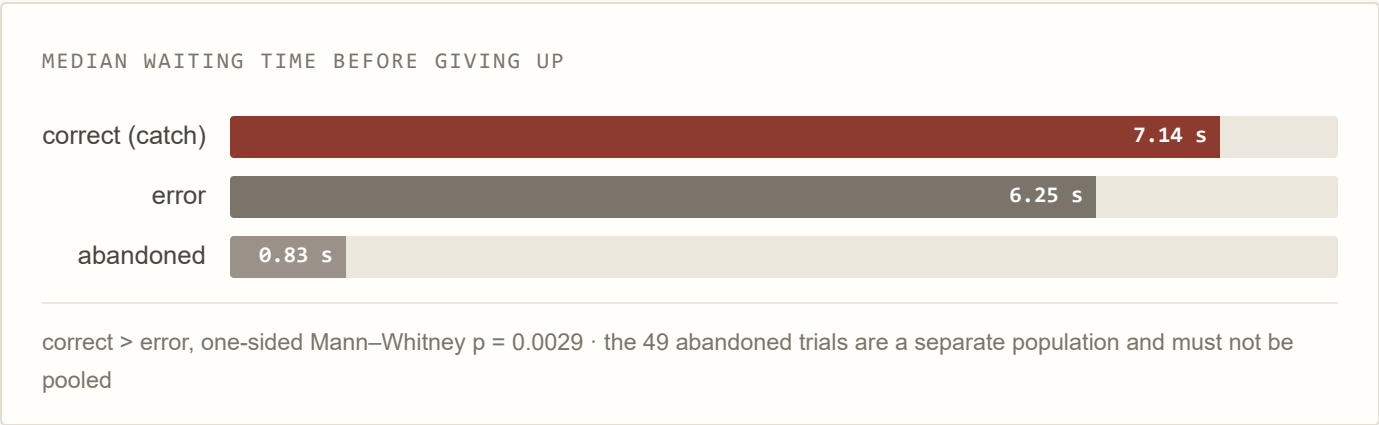
The three kinds of trial, and what the rat's patience reveals

Completed trials divide three ways, and the middle group is the informative one:

GROUP	N	WHAT HAPPENED
Error	71	chose the wrong side
Correct + rewarded	291	chose right, water arrived
Correct + omission	116	chose right, no water — 67 catch trials, 49 abandoned early

The 116 omission trials must be split before anything is measured — the two halves are different populations.

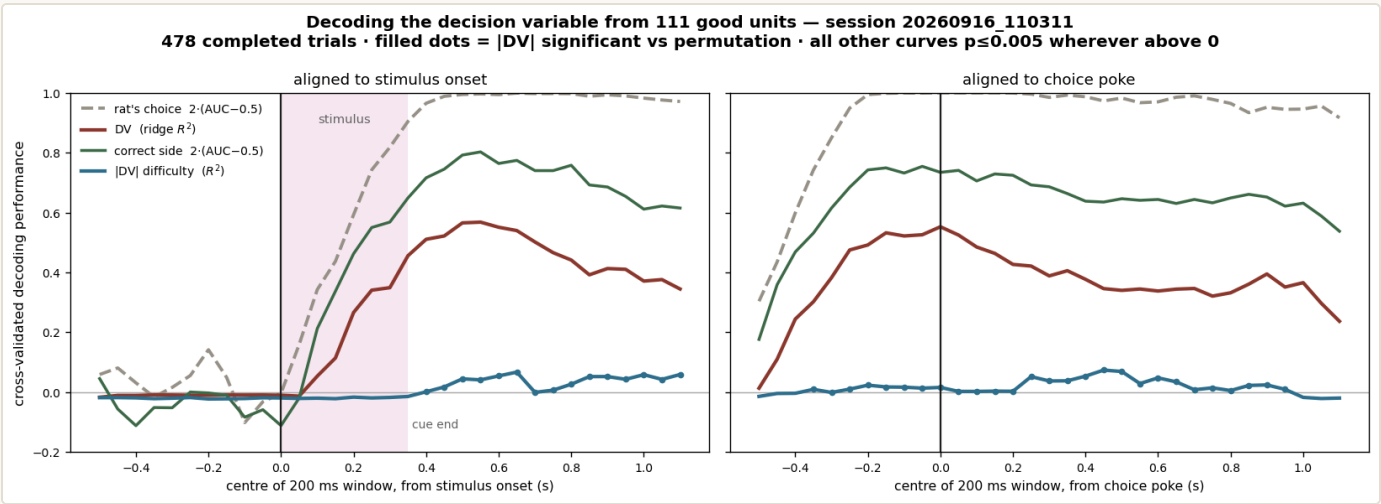
On an omission trial the rat cannot tell it from a rewarded trial until the water fails to come. **How long it holds on is a confidence report** — and because catch trials and error trials both have their wait stretched to 20 s, they compare directly:



And on catch trials the wait **scales with how easy the trial was** ($\rho = +0.330$, $p = 0.0064$; quartile medians 6.36 → 7.06 → 6.85 → 8.05 s). On error trials it does not ($\rho = +0.027$, $p = 0.83$). That last contrast is the clincher: **when the rat was wrong, its persistence stops tracking the true stimulus strength** — exactly what a confidence report should do.

Can the decision variable be decoded?

Yes — and the answer is more interesting than it first looks. 111 units, 478 trials, 200 ms sliding windows, ridge / linear SVR / random forest / logistic / LDA, every number against **its own permutation null**.



The rat's choice (grey, dashed) leads everything. DV and correct-side track it. Difficulty (blue) stays on the floor throughout — it is the only curve that is not a motor signal.

Before the stimulus: nothing, across every method — DV $R^2 = -0.009$ (ridge), -0.073 (random forest), difficulty $p = 0.47-0.80$, choice AUC 0.51. A clean negative control.

DV becomes decodable ~100 ms after stimulus onset and peaks at $R^2 = 0.568$. But three independent tests show it is the *choice*, not the stimulus:

TEST	RESULT	READING
Variance ceiling	choice alone explains $R^2 = 0.533$	decoder reached 0.564 — nothing beyond choice
AUC on error trials	0.002 / 0.001 / 0.003	inverted, not chance — it predicts the choice

Random forest vs
ridge

0.240 vs 0.251

no hidden nonlinear structure

Train sign(DV) on the 407 correct trials, test on the 71 errors — where the rat's choice and the correct side disagree.

The one genuinely non-motor signal is |DV| — difficulty, which is independent of which side was correct:

EPOCH	RIDGE R^2	FOREST R^2	P
pre-stimulus	-0.023	-0.045	0.47
during stimulus	-0.018	-0.027	0.17
cue end → choice	+0.039	+0.091	0.008
at choice poke	+0.011	+0.060	0.008
after choice (wait)	+0.083	+0.068	0.008
late wait	+0.011	+0.035	0.008

Difficulty decoding, ridge and random forest, each against its own permutation null.

Pooled across all trials, **difficulty carries no information while the stimulus is playing** — not even nonlinearly — and appears only after the choice, peaking ~450–500 ms into the waiting period, exactly where the confidence behaviour sits. Difficulty also predicts how long the rat waits ($\rho = +0.33$) and whether it was right (mean |DV| 0.533 on correct trials vs **0.208** on errors), so that post-choice signal reads as **the rat's confidence state** rather than a sensory one.

But pooling hid something. Decoding *within* a single choice — where choice is constant, so nothing decodable can be motor — difficulty **is** present during the stimulus on right-choice trials ($R^2 = +0.077$, $p = 0.005$), rising to +0.187 at cue end. Pooling cancels it, which is what happens when the code is choice-dependent. The effect is one-sided (right is contralateral to this animal's left-hemisphere implant, the expected direction for striatum) but n is also larger there (270 vs 208), so power isn't controlled. Stimulus information is present — small, and organised per choice.

The limit no analysis here can get past: choice is decodable at AUC 0.67 within 100 ms of stimulus onset and 0.80 by 200 ms. The decision forms while the tone is still playing, so no

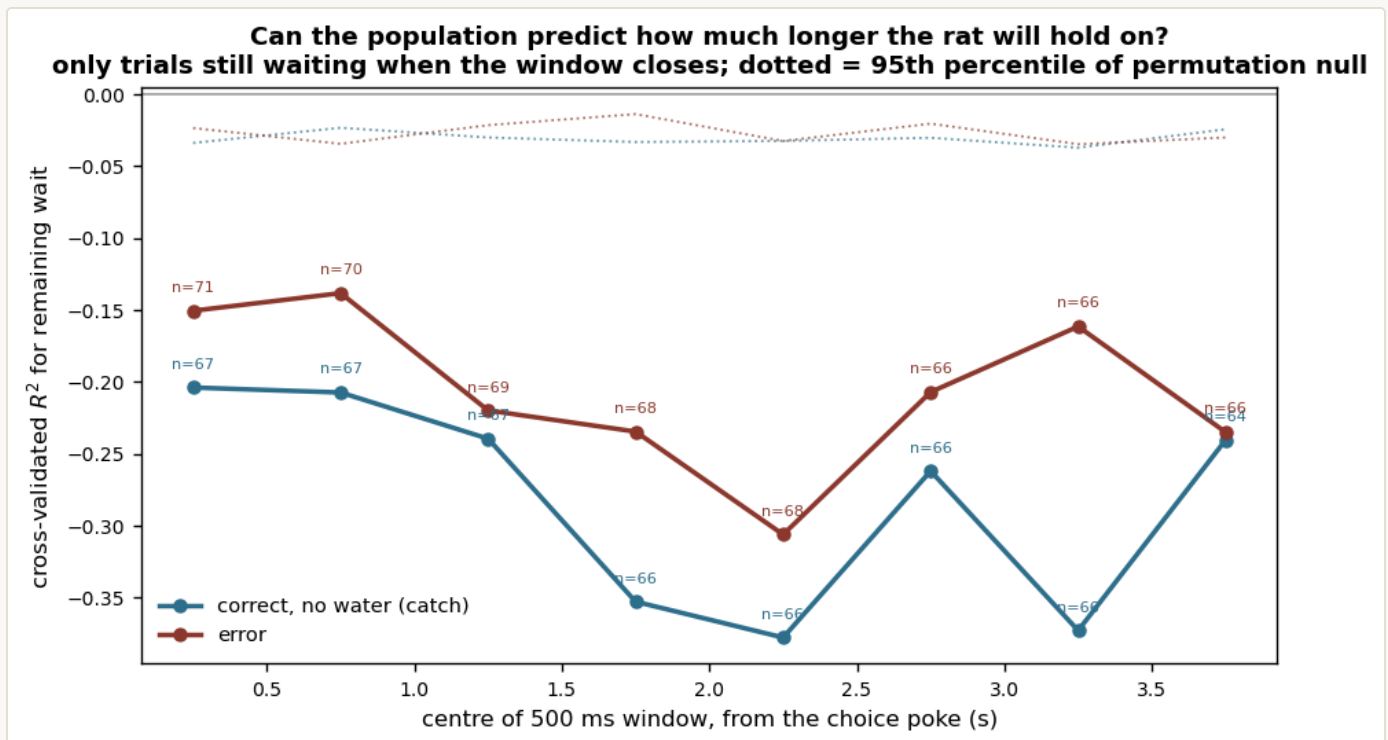
window in this task separates sensory coding from the developing choice. That needs a design where the choice is withheld or delayed.

When does the rat decide to give up?

On trials where no water ever comes, the rat's departure is its own decision — so it can be predicted, or not. Rewarded trials are excluded (the valve ends those, not the rat), errors and omissions kept separate.

The confound that would otherwise fake this. A fixed window 3 s after the choice poke, applied to every trial, measures activity *after the rat has already left* on short-wait trials — so a "decoder" would really be detecting whether it's still in the port. Every window below includes only trials still waiting when it closes. This is also why long-wait trials are the only well-posed regime.

How much longer will it stay? Not decodable.



Cross-validated R^2 for remaining wait, every lag from 0 to 4 s. Both groups sit well below their permutation nulls (dotted) — fitting noise, not signal.

R^2 is **negative at every lag in both groups** (catch -0.20 to -0.38 , error -0.14 to -0.31) against nulls near -0.03 , $p = 0.54$ – 0.87 . There is no graded countdown. With $n = 64$ – 71 per group and only 1.6 – 1.9 s of SD in remaining wait, this is a low-powered negative rather than a proof of absence.

Is the leave imminent? Strongly yes.

GROUP	TRIALS	TIME POINTS	AUC	CHANCE (95TH)	P
Correct, no water (catch)	67	1,878	0.923	0.766 (0.811)	0.005
Error	69	1,715	0.901	0.769 (0.809)	0.005

*From a 300 ms window, will the rat go within the next 500 ms? Cross-validated by trial. The null shuffles which trial got which leave time, so it **preserves elapsed time as a predictor** — chance here is 0.77, not 0.5.*

The dissociation is the result. Nothing predicts *how much longer* the rat will hold on — but the moment of leaving is flagged about half a second ahead, well beyond what elapsed time alone provides. Both groups behave identically, so it is not specific to being right or wrong.

Hold the interpretation loosely: leaving is a movement, so a signal 500 ms ahead of it is at least partly motor preparation. Separating “decided to quit” from “about to withdraw” needs an independent readout of the decision, which this task doesn’t provide.

What the sorting quality means for these results

These responses were measured on hand-curated units from a sorter whose average cluster is 40% foreign. That sounds fatal and mostly is not, for a specific reason: contamination **adds** other neurons’ spikes to a cluster. If the contaminating neurons are not themselves reward- or cue-locked, they dilute a real response toward zero rather than inventing one. So the responses reported here are more likely *under*-stated than fabricated.

Two things it does cost. Effect sizes are attenuated by an unknown factor. And **side selectivity is the vulnerable measurement** — if a contaminating neighbour has its own side preference, it can blunt or distort a unit’s. Once footprint-first attribution exists, the 45 side-selective units are the first result to re-measure.

Session 20260916_110311 · rat Shamir · 953 trials / 478 completed · 448 clusters / 111 hand-curated good
Benchmark: 33 hand-reviewed donor units, 52 placements, 15,086 ground-truth spikes, 4 replicates, 9
configurations

Code and full record: [post_ks_correction/RESEARCH_LOG.md](#) §5ak-5an · commits 012386e, ebd0747, aab372e, 03eeb07